keke_model

```
#load packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
#load data
train <- read_csv("data-train.csv")
```

```
Rows: 89 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (7): St, Re, Fr, R_moment_1, R_moment_2, R_moment_3, R_moment_4

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#transform data
#making new variables for our summary stats

#mean = R_moment_1, C_moment_2 is variance, everything else is as commented
train <- train %>%
```

```
  mutate(
    m1 = R_moment_1, m2 = R_moment_2, m3 = R_moment_3, m4 = R_moment_4,

    # central moments
    C_moment_2 = m2 - m1^2,                                  # variance
    C_moment_3 = m3 - 3*m1*m2 + 2*m1^3,
    C_moment_4 = m4 - 4*m1*m3 + 6*m1^2*m2 - 3*m1^4,

    # standardized versions
    sd_R     = sqrt(C_moment_2),
    skewness = C_moment_3 / sd_R^3,
    kurtosis = C_moment_4 / sd_R^4,                          # raw kurtosis
    ex_kurt  = kurtosis - 3                                  # excess kurtosis
  )

train <- train %>%
  mutate(
    Re_group = factor(Re),
    Fr_group = factor(Fr)
  )

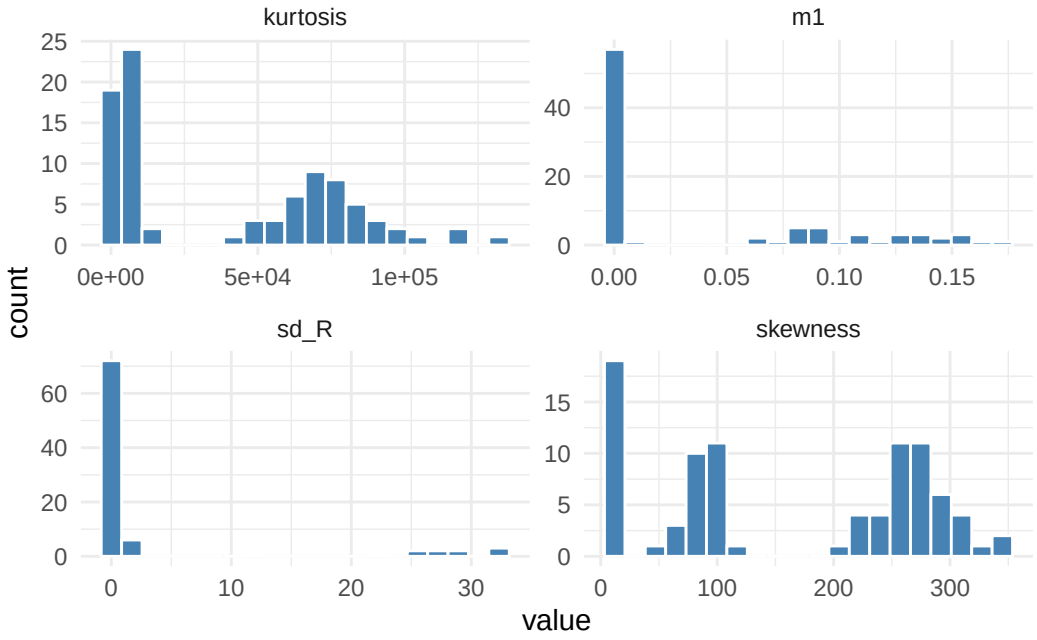
outcomes <- c("m1", "sd_R", "skewness", "kurtosis")

summary(train[outcomes])
```

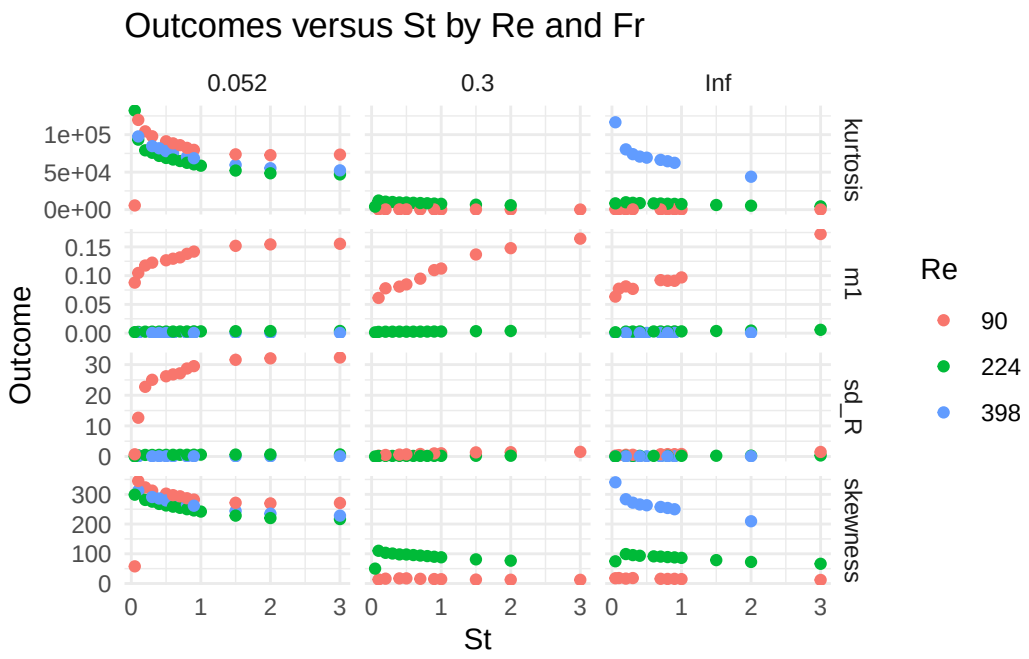
```
       m1                sd_R              skewness          kurtosis
 Min.   :0.000222   Min.   : 0.01006   Min.   : 11.97   Min.   :   150.5
 1st Qu.:0.002157   1st Qu.: 0.15643   1st Qu.: 72.55   1st Qu.:  5622.3
 Median :0.002958   Median : 0.28423   Median :110.12   Median : 12158.7
 Mean   :0.040394   Mean   : 3.65112   Mean   :162.81   Mean   : 39749.6
 3rd Qu.:0.087868   3rd Qu.: 0.72579   3rd Qu.:269.54   3rd Qu.: 72732.4
 Max.   :0.172340   Max.   :32.31526   Max.   :344.91   Max.   :132136.7
```

```
eda_data <- train %>%
  select(St, Re_group, Fr_group, all_of(outcomes)) %>%
  pivot_longer(
    cols = all_of(outcomes),
    names_to = "outcome",
    values_to = "value"
  )
```

```r
# Distributions of the outcomes
ggplot(eda_data, aes(x = value)) +
  geom_histogram(bins = 20, color = "white", fill = "steelblue") +
  facet_wrap(~ outcome, scales = "free") +
  theme_minimal()
```




```r
# Relationships with St, separated by Re and Fr
ggplot(
  eda_data,
  aes(x = St, y = value, color = Re_group)
) +
  geom_point() +
  facet_grid(outcome ~ Fr_group, scales = "free_y") +
  labs(color = "Re", y = "Outcome", title = "Outcomes versus St by Re and Fr") +
  theme_minimal()
```




```
results <- list()
fits <- list()

for (outcome in outcomes) {
  for (degree in 1:5) {

    model_formula <- as.formula(
      paste0(
        outcome,
        " ~ Re_group + Fr_group + poly(St, ",
        degree, ")"
      )
    )

    predictions <- numeric(nrow(train))

    # Leave-one-out cross-validation
    for (i in seq_len(nrow(train))) {
      fit <- lm(model_formula, data = train[-i, ])

      predictions[i] <- predict(
        fit,
        newdata = train[i, , drop = FALSE]
```

```
      )
    }

    observed <- train[[outcome]]
    model_name <- paste(outcome, degree, sep = "_degree_")

    results[[model_name]] <- data.frame(
      outcome = outcome,
      degree = degree,
      RMSE = sqrt(mean((observed - predictions)^2)),
      MAE = mean(abs(observed - predictions))
    )

    # Save the model fitted on all observations
    fits[[model_name]] <- lm(model_formula, data = train)
  }
}

cv_results <- do.call(rbind, results)
rownames(cv_results) <- NULL

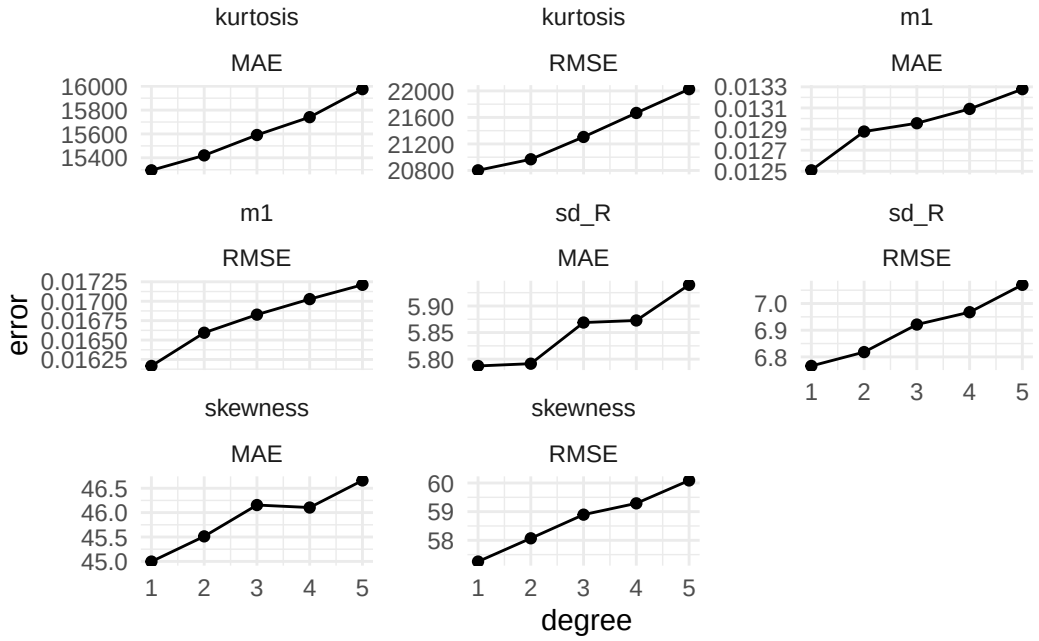
cv_results <- cv_results[
  order(cv_results$outcome, cv_results$RMSE),
]

print(cv_results)
```

```
    outcome degree         RMSE          MAE
16 kurtosis      1 2.080217e+04 1.529544e+04
17 kurtosis      2 2.096850e+04 1.542056e+04
18 kurtosis      3 2.130548e+04 1.559127e+04
19 kurtosis      4 2.166752e+04 1.574075e+04
20 kurtosis      5 2.202634e+04 1.597525e+04
1        m1      1 1.616672e-02 1.250941e-02
2        m1      2 1.659411e-02 1.287653e-02
3        m1      3 1.682715e-02 1.295516e-02
4        m1      4 1.702713e-02 1.309085e-02
5        m1      5 1.721137e-02 1.327776e-02
6      sd_R      1 6.766073e+00 5.787222e+00
7      sd_R      2 6.817985e+00 5.791457e+00
8      sd_R      3 6.921242e+00 5.868843e+00
9      sd_R      4 6.967510e+00 5.872705e+00
```

```
10      sd_R      5 7.070173e+00 5.939783e+00
11 skewness       1 5.725744e+01 4.499625e+01
12 skewness       2 5.806958e+01 4.551313e+01
13 skewness       3 5.889492e+01 4.615527e+01
14 skewness       4 5.928963e+01 4.610428e+01
15 skewness       5 6.008891e+01 4.665974e+01
```

```r
# Plot both error measures
cv_results %>%
  pivot_longer(
    cols = c(RMSE, MAE),
    names_to = "metric",
    values_to = "error"
  ) %>%
  ggplot(aes(x = degree, y = error)) +
  geom_line() +
  geom_point() +
  facet_wrap(vars(outcome, metric), scales = "free_y") +
  scale_x_continuous(breaks = 1:5) +
  theme_minimal()
```



```r
# Lowest LOOCV RMSE for each outcome
best_models <- cv_results %>%
  group_by(outcome) %>%
  slice_min(RMSE, n = 1, with_ties = FALSE) %>%
  ungroup()

best_models
```

```
# A tibble: 4 x 4
  outcome  degree       RMSE        MAE
  <chr>     <int>      <dbl>      <dbl>
1 kurtosis      1 20802.     15295.
2 m1            1     0.0162     0.0125
3 sd_R          1     6.77       5.79
4 skewness      1    57.3       45.0
```